FLORENT TARIOLLE

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Engineering student seeking a 6-month final-year research internship in deep learning, starting in 2027.

EDUCATION

INSA Rouen Normandy — Rouen, France
Engineering degree, Computer Science and AI, expected 2027 — 2023 - 2027

University of Rouen Normandy — Rouen, France
M.Sc. in Data Science, research-oriented MLAI track, expected 2027 — 2026 - 2027

Lycée Dupuy de Lôme — Lorient, France
CPGE MP (Preparatory Classes for Grandes Écoles, Mathematics and Physics) — 2021 - 2023

PUBLICATIONS

Opportunistic Target Selection — May 2026
F. Tariolle and F. Yger. *CAp 2026 (Conférence sur l'Apprentissage automatique).*

- Introduced a lightweight wrapper for score-based black-box adversarial attacks using early class confidence signals to reduce class drift in undirected random-search attacks.
- Improved attacks by up to 27 percentage points in success rate and a 43% relative reduction in censored-mean iterations on ResNet-50, validated across 4,500 ImageNet attack runs on five standard classifiers.

EXPERIENCE

InterDigital — Rennes, France
Deep Learning Research Intern — May 2026 - Present

- Working on deep learning methods for next-generation video compression, focusing on VVC (H.266) in-loop filtering and decoder-side filter selection to improve reconstructed video quality and coding efficiency.

Enedis — Rouen, France
Machine Learning Intern — Jan 2026 - May 2026

- Working on deep generative models and large-scale pipelines to simulate high-frequency time series for 38 million customers, supporting a digital twin of the French electricity grid.

RESEARCH PROJECTS

SLS-WM: Annealed Structured Label Smoothing for Discrete World Models — Feb 2026 - Present
Independent research project; manuscript in preparation.

- Designed a tokenizer-metric-aware objective for discrete world models that uses local soft targets early in training and anneals back to cross-entropy to preserve the final objective.
- Evaluating the method on an accepted IRIS Atari world-model baseline under matched cross-entropy vs. annealed-SLS settings, measuring optimization stability, sample efficiency, final/tail return, and seed failure rate.
- Built an action-conditioned Geometry Dash world model with FSQ tokenization, a block-causal dynamics transformer, and a latent actor-critic trained in imagined rollouts; deployed zero-shot on real gameplay at 30 FPS.

AWARDS

Finalist, Hack the World(s) Hackathon — Jun 2026
24-hour team project at a Yann LeCun-sponsored hackathon on JEPAs and world models

- Developed a geometry-aware joint-embedding EEG SSL system: strong SIGReg frozen-transfer baselines and Riemannian latent analysis showing structure in SPD covariance representations.

OPEN SOURCE

Rose (Active: 15K+ DAU) — Sep 2025 - Present
Co-founder & Lead Developer

- Co-founded and led development of a high-traffic real-time customization tool used by 15K+ daily active users; coordinated a 6-person team and designed the Python backend, Cloudflare Workers WebSocket relay, and browser-side JavaScript integration.
- Replaced OCR-based state extraction with deterministic DOM parsing, reducing latency and operational complexity.

TECHNICAL SKILLS

Programming: Python, C++
AI & ML: PyTorch, CUDA; Deep Learning, Self-Supervised Learning, World Models, Model-Based RL and Planning
Languages: French (native), English (TOEIC 970/990)

REFERENCES

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Frédéric Lefebvre — frederic.lefebvre@interdigital.com